From the LW Frobenius Risk to SURE Form

Setup

The shrinkage estimator is $\hat{\Sigma} = \alpha F + (1-\alpha)S$, so entrywise $\hat{\sigma}_{ij} = \alpha f_{ij} + (1-\alpha)s_{ij}$. The risk is the expected squared Frobenius norm,

$$R(\alpha) = \mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \sum_{i,j} \mathbb{E}(\hat{\sigma}_{ij} - \sigma_{ij})^2.$$

The only assumption needed is that S is unbiased: $\mathbb{E}\, s_{ij} = \sigma_{ij}$. Everything below is done for one fixed entry (i, j) and summed at the end.

Step 1: Lemma 1 (add and subtract S)

Write

$$\hat{\sigma}_{ij} - \sigma_{ij} = (\hat{\sigma}_{ij} - s_{ij}) + (s_{ij} - \sigma_{ij}).$$

Squaring,

$$(\hat{\sigma}_{ij} - \sigma_{ij})^2 = (\hat{\sigma}_{ij} - s_{ij})^2 + (s_{ij} - \sigma_{ij})^2 + 2(\hat{\sigma}_{ij} - s_{ij})(s_{ij} - \sigma_{ij}).$$

Take expectations of the three pieces.

Piece 1. Keep $\mathbb{E}(\hat{\sigma}_{ij} - s_{ij})^2$ as it is.

Piece 2. Since $\mathbb{E}\, s_{ij} = \sigma_{ij}$, $\mathbb{E}(s_{ij} - \sigma_{ij})^2 = \mathrm{Var}(s_{ij})$.

Piece 3 (cross term). Split it:

$$\mathbb{E}\big[(\hat{\sigma}_{ij} - s_{ij})(s_{ij} - \sigma_{ij})\big] = \mathbb{E}\big[\hat{\sigma}_{ij}(s_{ij} - \sigma_{ij})\big] - \mathbb{E}\big[s_{ij}(s_{ij} - \sigma_{ij})\big].$$

Because $s_{ij} - \sigma_{ij}$ has mean zero, for any random variable Y we have $\mathbb{E}[Y(s_{ij} - \sigma_{ij})] = \mathrm{Cov}(Y, s_{ij})$. Taking $Y = \hat{\sigma}_{ij}$ and then $Y = s_{ij}$:

$$\mathbb{E}\big[(\hat{\sigma}_{ij} - s_{ij})(s_{ij} - \sigma_{ij})\big] = \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij}) - \mathrm{Var}(s_{ij}).$$

Combine.

$$\begin{aligned}\mathbb{E}(\hat{\sigma}_{ij} - \sigma_{ij})^2 &= \mathbb{E}(\hat{\sigma}_{ij} - s_{ij})^2 + \mathrm{Var}(s_{ij}) + 2\,\mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij}) - 2\,\mathrm{Var}(s_{ij}) \\ &= \mathbb{E}(\hat{\sigma}_{ij} - s_{ij})^2 - \mathrm{Var}(s_{ij}) + 2\,\mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij}).\end{aligned}$$

Summing over i, j gives Lemma 1 of Yi and Zou (2013):

$$\boxed{\mathbb{E}\|\hat{\Sigma} - \Sigma\|_F^2 = \mathbb{E}\|\hat{\Sigma} - S\|_F^2 - \sum_{i,j} \mathrm{Var}(s_{ij}) + 2\sum_{i,j} \mathrm{Cov}(\hat{\sigma}_{ij}, s_{ij})}$$

This holds for *any* estimator $\hat{\Sigma}$; no normality is used.

Step 2: Plug in the LW shrinker

First term. $\hat{\sigma}_{ij} - s_{ij} = \alpha f_{ij} + (1-\alpha)s_{ij} - s_{ij} = \alpha(f_{ij} - s_{ij})$, so

$$\mathbb{E}\|\hat{\Sigma} - S\|_F^2 = \alpha^2\, \mathbb{E}\|F - S\|_F^2.$$

Covariance term. Covariance is linear in each argument, so

$$\operatorname{Cov}(\hat{\sigma}_{ij}, s_{ij}) = \operatorname{Cov}\left(\alpha f_{ij} + (1-\alpha)s_{ij},\, s_{ij}\right) = \alpha \operatorname{Cov}(f_{ij}, s_{ij}) + (1-\alpha)\operatorname{Var}(s_{ij}).$$

Collect the variance terms. The Var terms are $-\operatorname{Var}(s_{ij}) + 2(1-\alpha)\operatorname{Var}(s_{ij}) = (1-2\alpha)\operatorname{Var}(s_{ij})$.

Result

$$R(\alpha) = \underbrace{\alpha^2\, \mathbb{E}\|F - S\|_F^2}_{\text{computable from data}} + (1-2\alpha) \underbrace{\sum_{i,j} \operatorname{Var}(s_{ij})}_{\text{estimated by Lemma 2}} + 2\alpha \underbrace{\sum_{i,j} \operatorname{Cov}(f_{ij}, s_{ij})}_{\text{unknown}}.$$

Remark: why $\mathbb{E}\, s_{ij} = \sigma_{ij}$ matters

If S divides by T and the mean is estimated, then $\mathbb{E}\, S = \frac{T-1}{T}\Sigma \neq \Sigma$. Piece 2 then becomes $\operatorname{Var}(s_{ij}) + b_{ij}^2$ with bias $b_{ij} = \mathbb{E}s_{ij} - \sigma_{ij}$, and the cross term picks up an extra $-2\, b_{ij}\, \mathbb{E}(\hat{\sigma}_{ij} - s_{ij})$ term. Using S with divisor $T-1$ (or a known zero mean) avoids this.